Chapter 10 — Basic Electrical Engineering

Syllabus (CEN 05/2025, Annexure-VII C, Unit 10): *DC Circuits; AC fundamentals; Magnetic, Thermal and Chemical effects of Electric current; Earthing — Installation, Maintenance, Testing.*

10.0 Weightage & Trend Box

Item	Detail
Weight in CBT-2	~8% of technical marks (4 of 50 technical Q in the 22-04-2025 Shift-2 paper)
Difficulty	Easy–moderate; mostly conceptual one-liners + light numericals
Special trend	The 2025 paper pulled **protection-relay** questions into this unit — so relays/protection are covered here even though the syllabus line does not name them

Actual questions observed in the 22-04-2025 CBT-2 paper (this unit):

1. **Relay selectivity** → the quality by which a relay operates **only for faults in its own designated (protected) zone** and stays inoperative for faults outside it.
2. **Distance relay** → operates on the **measurement of impedance (ratio V/I)** at the relay location, which is proportional to the distance to the fault.
3. **Magnetic circuit** → for the same MMF, **increased reluctance → decreased flux** (Φ = MMF/S).
4. **Magnetic field** → the **region around a magnet (or current-carrying conductor)** in which its magnetic force can be experienced.

Preparation strategy: master (a) series–parallel + KCL/KVL numericals, (b) RMS/average/form-factor values, (c) RLC impedance & resonance formulas, (d) the electric–magnetic circuit analogy table, (e) relay qualities and one-line working of each relay type, (f) plate/pipe earthing dimensions & permissible earth-resistance values. These six clusters cover essentially every question RRB has asked from this unit.

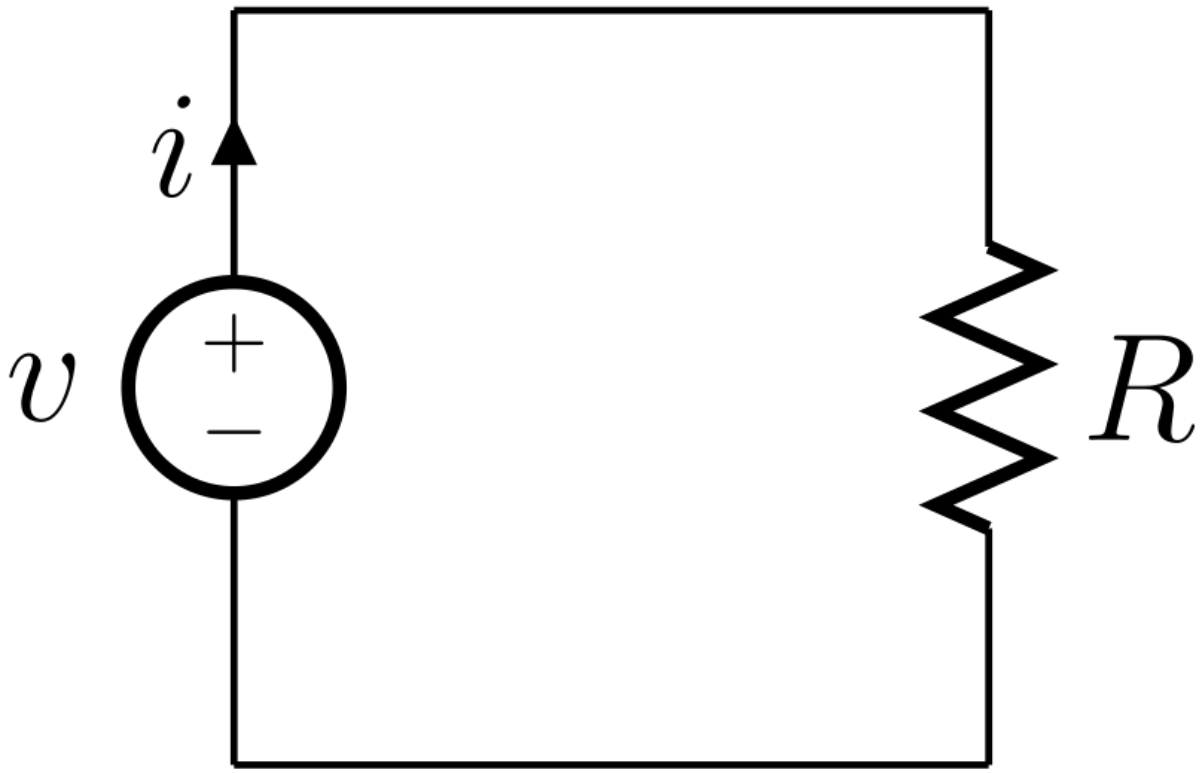
10.1 DC Circuits

10.1.1 Charge, Current, Voltage, Power, Energy

Quantity	Symbol	Definition	SI Unit
Charge	Q	Deficiency/excess of electrons; Q = It	coulomb (C); 1 C = charge of 6.24×10^{18} electrons
Current	I	Rate of flow of charge, I = dQ/dt	ampere (A) = C/s
Voltage (potential difference)	V	Work done per unit charge, V = W/Q	volt (V) = J/C
EMF	E	Energy given per unit charge by a source	volt (V)
Power	P	Rate of doing work, $P = VI = I^2R = V^2/R$	watt (W) = J/s
Energy	W	W = Pt = VIt	joule (J); commercial unit: 1 kWh = 3.6×10^6 J

- Conventional current flows from + to − outside... no — **outside the source, conventional current flows from + terminal through the external circuit to − terminal**; electron flow is opposite.
- 1 hp = 746 W (British) — frequently asked.

10.1.2 Ohm's Law and Its Limitations



Ohm's-law circuit.

Statement: At constant temperature and other physical conditions, the current through a conductor is directly proportional to the potential difference across it.

> $V = IR \Rightarrow R = V/I$ (ohm, Ω)

Limitations (non-ohmic cases):

- Not valid for **non-linear** devices: diodes, transistors, vacuum tubes, thyristors.
- Not valid for **unilateral** elements (conduct differently in the two directions).
- Not valid when temperature changes appreciably (filament lamp), for arc discharge, electrolytes with gas evolution, semiconductors.

10.1.3 Resistance — Laws, Factors, Temperature Coefficient

Laws of resistance: $R = \rho l/A$

Factor	Effect
Length l	$R \propto l$
Area A	$R \propto 1/A$
Material	via resistivity ρ (Ω·m)
Temperature	metals: R increases; semiconductors/insulators/electrolytes/carbon: R decreases

- Conductance $G = 1/R$ (siemens, S or ℧); conductivity $\sigma = 1/\rho$.
- Resistivity of copper ≈ 1.72×10^{-8} Ω·m; of aluminium ≈ 2.8×10^{-8} Ω·m.

Temperature coefficient of resistance (α):

> $R_2 = R_1[1 + \alpha_1(t_2 - t_1)]$

α_0 for copper ≈ 0.00427 /°C at 0 °C (≈ 1/234.5). Also $\alpha_2 = \alpha_1 / [1 + \alpha_1(t_2 - t_1)]$ and $\alpha_1 = \alpha_0/(1 + \alpha_0 t_1)$.

- α positive → metals; α negative → carbon, semiconductors, insulators, electrolytes.
- Alloys (manganin, constantan, eureka) have α ≈ 0 → used for standard resistors and shunts.

Worked Example 1. A copper coil has resistance 40 Ω at 20 °C. Find its resistance at 60 °C. (α at 20 °C = 0.004 /°C)

$R_{60} = R_{20}[1 + \alpha(60 - 20)] = 40[1 + 0.004 \times 40] = 40 \times 1.16 =$ **46.4 Ω**

Worked Example 2. A filament's resistance at 0 °C is 100 Ω, $\alpha_0 = 0.005$/°C. At what temperature is the resistance 150 Ω?

$150 = 100(1 + 0.005\ t) \Rightarrow 0.005\ t = 0.5 \Rightarrow$ **t = 100 °C**

10.1.4 Series, Parallel and Series–Parallel Networks

Property	Series	Parallel
Equivalent R	$R = R_1 + R_2 + \ldots$	$1/R = 1/R_1 + 1/R_2 + \ldots$
Same quantity	Current	Voltage
Divides	Voltage ($\propto R$)	Current ($\propto 1/R$)
Two resistors	$R_1 + R_2$	$R_1R_2/(R_1 + R_2)$ (product/sum)
n equal resistors r	nr	r/n

- n equal resistors: (series value)/(parallel value) = n^2.
- Adding resistance in parallel always **reduces** the equivalent below the smallest branch.

Voltage divider (two resistors in series across V):

> $V_1 = V \cdot R_1/(R_1 + R_2)$, $V_2 = V \cdot R_2/(R_1 + R_2)$

Current divider (two resistors in parallel fed by I):

> $I_1 = I \cdot R_2/(R_1 + R_2)$, $I_2 = I \cdot R_1/(R_1 + R_2)$ (current takes the *other* branch's resistance in the numerator)

Worked Example 3. Three resistors 2 Ω, 3 Ω and 6 Ω are in parallel, and this combination is in series with 4 Ω across a 12 V battery. Find the battery current and the voltage across the parallel group.

$1/R_p = 1/2 + 1/3 + 1/6 = 1 \Rightarrow R_p = 1\ \Omega$. Total $R = 4 + 1 = 5\ \Omega$.

$I = 12/5 =$ **2.4 A**. Voltage across parallel group $= 2.4 \times 1 =$ **2.4 V**.

Worked Example 4 (current divider). A 10 A current divides between 6 Ω and 4 Ω in parallel.

$I_6 = 10 \times 4/(6+4) =$ **4 A**; $I_4 = 10 \times 6/10 =$ **6 A**. (Larger current through smaller resistance.)

10.1.5 Kirchhoff's Laws

KCL (Kirchhoff's Current Law / point law): The algebraic sum of currents meeting at a junction is zero — $\Sigma I = 0$ (based on **conservation of charge**). Currents in = currents out.

KVL (Kirchhoff's Voltage Law / mesh law): Around any closed loop, the algebraic sum of EMFs equals the algebraic sum of IR drops — $\Sigma E = \Sigma IR$ (based on **conservation of energy**).

Sign convention: going through a resistor **in the direction of current** → voltage drop (−IR); through a source from − to + → rise (+E).

Worked Example 5 (two-mesh problem). Two batteries $E_1 = 12$ V (internal $r_1 = 1\ \Omega$) and $E_2 = 8$ V ($r_2 = 1\ \Omega$) are connected in parallel, feeding a load $R = 6\ \Omega$. Find the load current.

Let I_1, I_2 be battery currents; load current $I = I_1 + I_2$, load voltage V.

Loop 1: $12 = 1 \cdot I_1 + 6(I_1 + I_2)$

Loop 2: $8 = 1 \cdot I_2 + 6(I_1 + I_2)$

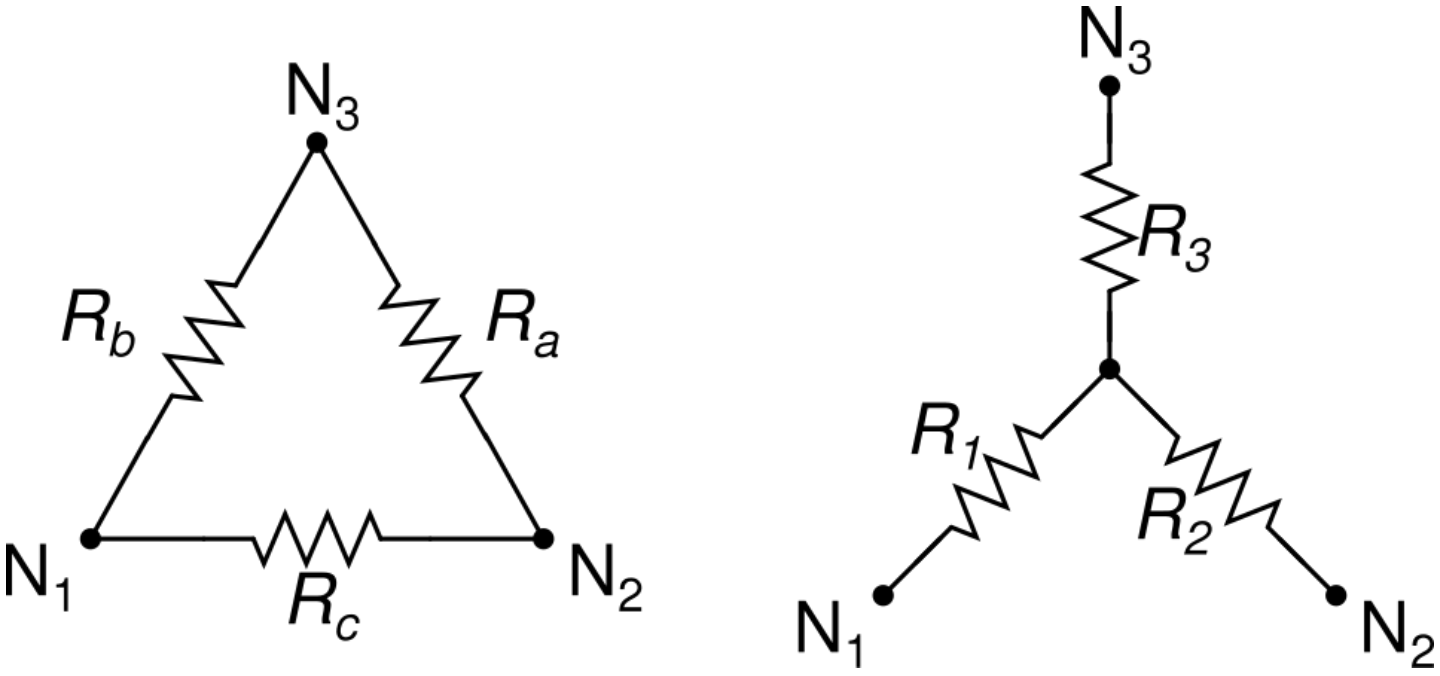
$\Rightarrow 7I_1 + 6I_2 = 12$ … (i); $6I_1 + 7I_2 = 8$ … (ii)

(i)×7 − (ii)×6: $49I_1 + 42I_2 - 36I_1 - 42I_2 = 84 - 48 \Rightarrow 13I_1 = 36 \Rightarrow I_1 = 2.769$ A

From (i): $6I_2 = 12 - 19.385 = -7.385 \Rightarrow I_2 = -1.231$ A (E_2 is being charged).

Load current I = 2.769 − 1.231 = **1.538 A**; load voltage = 6 × 1.538 = **9.23 V**.

10.1.6 Star–Delta (Y–Δ) Transformation



Star (Y) and delta (Δ) connections.

Delta → Star (each star arm = product of the two delta sides meeting at that terminal ÷ sum of all three):

> $R_1 = R_{12}R_{31}/(R_{12} + R_{23} + R_{31})$, and cyclically for R_2, R_3.

Star → Delta (each delta side = sum of the two star arms it connects + their product ÷ third arm):

> $R_{12} = R_1 + R_2 + R_1R_2/R_3$, and cyclically.

- If all three delta resistors are R, each star resistor = **R/3**. If all star resistors are R, each delta resistor = **3R**.

Worked Example 6. A delta has 6 Ω, 6 Ω, 6 Ω. Equivalent star arm = 6×6/(6+6+6) = 36/18 = **2 Ω** (= 6/3 ✓).

Worked Example 7. Star arms R_1 = 2 Ω, R_2 = 3 Ω, R_3 = 6 Ω. Find delta side between terminals 1 and 2.

R_{12} = 2 + 3 + (2×3)/6 = 5 + 1 = **6 Ω**. (Similarly R_{23} = 3 + 6 + 18/2 = 18 Ω; R_{31} = 6 + 2 + 12/3 = 12 Ω.)

10.1.7 Network Theorems

Superposition theorem. In a linear network with several sources, the current in any element = algebraic sum of currents produced by each source acting alone, other **voltage sources replaced by short circuits** and **current sources by open circuits** (internal resistances retained). Valid only for **linear** networks; not applicable to power (power is non-linear in I).

Thevenin's theorem. Any linear two-terminal network can be replaced by a single voltage source V_Th (open-circuit voltage at the terminals) in series with R_Th (resistance seen from the terminals with all internal sources killed).

> I_load = V_Th / (R_Th + R_L)

Norton's theorem. The same network can be replaced by a current source I_N (short-circuit current at the terminals) in parallel with R_N (= R_Th). Duality: V_Th = I_N R_N.

Maximum Power Transfer theorem. Maximum power is delivered to the load when **R_L = R_Th** (DC) or **Z_L = Z_Th*** (complex conjugate, AC).

> P_max = $V_{Th}^2/(4R_{Th})$; efficiency at maximum power transfer = **50%**.

Worked Example 8 (Thevenin). A 10 V source with 2 Ω internal resistance feeds a 3 Ω resistor; a load R_L = 5 Ω is connected across the 3 Ω. Find load current by Thevenin.

Remove R_L. V_Th = voltage across 3 Ω = 10 × 3/(2+3) = 6 V.

R_Th = 3 ∥ 2 = 1.2 Ω.

I_L = 6/(1.2 + 5) = 6/6.2 = **0.968 A**.

Worked Example 9 (Maximum power). For the same Thevenin equivalent (V_Th = 6 V, R_Th = 1.2 Ω), maximum power to a matched load:

R_L = 1.2 Ω; $P_{max} = 6^2/(4 \times 1.2) = 36/4.8 =$ **7.5 W**.

Worked Example 10 (Norton). For Example 8's network: I_N = short-circuit current = V_Th/R_Th = 6/1.2 = **5 A**, R_N = 1.2 Ω. Load current = I_N × R_N/(R_N + R_L) = 5 × 1.2/6.2 = 0.968 A ✓ (matches Thevenin).

Worked Example 11 (Superposition). Two sources: 10 V in series with 2 Ω on the left, 5 V in series with 3 Ω on the right, both feeding a common 6 Ω load between the junction and the bottom rail. Find load current.

10 V alone (5 V shorted): R seen by 10 V = 2 + (3∥6) = 2 + 2 = 4 Ω; source current = 2.5 A; load current I′ = 2.5 × 3/(3+6) = 0.833 A (downward).

5 V alone (10 V shorted): R = 3 + (2∥6) = 3 + 1.5 = 4.5 Ω; source current = 1.111 A; load current I″ = 1.111 × 2/(2+6) = 0.278 A (downward).

Total load current = 0.833 + 0.278 = **1.111 A** (≈ 10/9 A).

10.2 AC Fundamentals

10.2.1 Generation of Sinusoidal EMF

A coil rotating with angular velocity ω in a uniform magnetic field B has flux linkage NΦ_m cos ωt, so induced EMF:

> e = E_m sin ωt, where E_m = NBAω (N turns, area A)

- One full rotation of the coil in a 2-pole field = one cycle.
- Frequency of alternator: **f = PN/120** (P = poles, N = rpm). In India, supply frequency = **50 Hz**, standard domestic voltage 230 V (RMS).

10.2.2 Waveform Terms

Term	Meaning
Cycle	One complete set of positive + negative alternations
Time period T	Time for one cycle; T = 1/f
Frequency f	Cycles per second (hertz, Hz)
Amplitude (peak value) E_m	Maximum value attained
Instantaneous value	Value at a particular instant, $e = E_m \sin \omega t$
Angular frequency ω	$\omega = 2\pi f$ rad/s (at 50 Hz, ω = 314 rad/s)
Phase	Angular position within the cycle; phase difference = angle between two waveforms

10.2.3 Average and RMS Values (Sine Wave Derivations)

Average value (over half cycle, since full-cycle average of a sine = 0):

> $I_{av} = (1/\pi) \int_0^{\pi} I_m \sin \theta \, d\theta = (I_m/\pi)[-\cos \theta]_0^{\pi} = 2I_m/\pi =$ **$0.637\, I_m$**

RMS (effective) value — that steady DC which produces the same heating:

> $I_{rms} = \sqrt{[(1/2\pi) \int_0^{2\pi} I_m^2 \sin^2\theta \, d\theta]} = I_m/\sqrt{2} =$ **$0.707\, I_m$**

Form factor = RMS/Average = 0.707/0.637 = **1.11** (sine).

Peak (crest) factor = Peak/RMS = √2 = **1.414** (sine).

Waveform	RMS	Average	Form factor	Peak factor
Sine	$I_m/\sqrt{2}$	$2I_m/\pi$	1.11	1.414
Square	I_m	I_m	1.0	1.0
Triangular	$I_m/\sqrt{3}$	$I_m/2$	1.155	1.732
Half-wave rectified sine	$I_m/2$	I_m/π	1.57	2.0
Full-wave rectified sine	$I_m/\sqrt{2}$	$2I_m/\pi$	1.11	1.414

- Moving-iron / hot-wire instruments read **RMS**; PMMC reads **average (zero for pure AC)**; rectifier instruments are calibrated in RMS assuming sine (form factor 1.11).

Worked Example 12. A sinusoidal voltage is v = 311 sin 314t. Find RMS value, frequency and time period.

$V_{rms} = 311/\sqrt{2} \approx$ **220 V**; $\omega = 314 \Rightarrow f = 314/2\pi =$ **50 Hz**; $T = 1/50 =$ **20 ms**.

10.2.4 Phasors

A phasor is a rotating line whose length = peak (or RMS) value and whose angular velocity = ω; its projection generates the sine wave. Phasors let us add/subtract sinusoids of the **same frequency** as vectors. j-operator: multiplication by j = 90° anticlockwise rotation; $j^2 = -1$.

10.2.5 Pure R, L, C in AC Circuits

Element	Voltage–current phase	Opposition	Power
R	In phase ($\varphi = 0$)	R (independent of f)	$P = VI = I^2R$
L	Current **lags** voltage by 90°	$X_L = \omega L = 2\pi fL$ ($\propto f$)	Average power = 0
C	Current **leads** voltage by 90°	$X_C = 1/\omega C = 1/(2\pi fC)$ ($\propto 1/f$)	Average power = 0

Mnemonic: **CIVIL** — in C, I leads V; V leads I in L.

- At DC (f = 0): inductor = short circuit, capacitor = open circuit.
- At very high frequency: inductor → open, capacitor → short.

10.2.6 RL, RC, RLC Series Circuits

Impedance Z (Ω) is the total opposition; phasor sum of R and net reactance.

Circuit	Impedance	Phase angle φ	Nature
R–L series	$Z = \sqrt{(R^2 + X_L^2)}$	$\tan\varphi = X_L/R$ (current lags)	Lagging pf
R–C series	$Z = \sqrt{(R^2 + X_C^2)}$	$\tan\varphi = X_C/R$ (current leads)	Leading pf
R–L–C series	$Z = \sqrt{[R^2 + (X_L - X_C)^2]}$	$\tan\varphi = (X_L - X_C)/R$	Lags if $X_L > X_C$; leads if $X_C > X_L$

Voltage triangle: $V = \sqrt{(V_R^2 + (V_L - V_C)^2)}$. Power factor **$\cos\varphi = R/Z = V_R/V = P/S$**.

Worked Example 13. A coil of R = 6 Ω, L = 25.5 mH is connected to 230 V, 50 Hz. Find current, power factor and power.

$X_L = 2\pi \times 50 \times 0.0255 \approx 8\ \Omega$. $Z = \sqrt{(6^2 + 8^2)} = 10\ \Omega$.

$I = 230/10 =$ **23 A**; $\cos\varphi = 6/10 =$ **0.6 lagging**; $P = VI\cos\varphi = 230 \times 23 \times 0.6 =$ **3174 W** ($= I^2R = 23^2 \times 6$ ✓).

Worked Example 14 (RLC series). R = 10 Ω, X_L = 30 Ω, X_C = 20 Ω on 200 V supply.

$Z = \sqrt{(10^2 + (30-20)^2)} = \sqrt{200} = 14.14\ \Omega$; $I = 200/14.14 =$ **14.14 A**;

$\varphi = \tan^{-1}(10/10) =$ **45° lagging**; $P = I^2R = 200 \times 10 / 200 \rightarrow I^2R = (14.14)^2 \times 10 =$ **2000 W**.

10.2.7 Power in AC Circuits — Power Triangle

Power	Formula	Unit
Active (true/real) P	$VI\cos\varphi = I^2R$	watt (W)
Reactive Q	$VI\sin\varphi = I^2X$	volt-ampere reactive (VAR)
Apparent S	$VI = I^2Z$	volt-ampere (VA)

> $S^2 = P^2 + Q^2$; power factor $\cos\varphi = P/S$.

- Power factor range 0 (pure L or C) to 1 (pure R). Low pf ⇒ higher current for same P ⇒ larger losses, poorer regulation, bigger equipment. Improvement: **static capacitors** in parallel (also synchronous condensers, phase advancers).
- Wattmeter reads P; energy meter reads kWh.

10.2.8 Resonance

Series resonance (acceptor circuit): $X_L = X_C \Rightarrow$

> $f_r = 1/(2\pi\sqrt{(LC)})$

At resonance: Z = R (minimum), I = V/R (maximum), pf = 1, $V_L = V_C$ (may far exceed supply voltage — voltage magnification).

Quality factor: $Q = X_L/R = \omega L/R = (1/R)\sqrt{(L/C)} = f_r/BW$ = voltage across L (or C) ÷ supply voltage.

Bandwidth: $BW = f_2 - f_1 = f_r/Q = R/(2\pi L)$; half-power points where I = 0.707 I_{max}.

Parallel resonance (rejector / tank circuit): coil (R, L) parallel with C:

> $f_r = (1/2\pi)\sqrt{(1/LC - R^2/L^2)} \approx 1/(2\pi\sqrt{(LC)})$ for small R

At resonance: impedance is **maximum** = dynamic impedance **$Z_d = L/CR$** (purely resistive), line current **minimum**, pf = 1; large circulating current between L and C — hence "tank circuit" (stores oscillating energy; used in oscillators, tuning).

Worked Example 15. A series circuit has R = 10 Ω, L = 100 mH, C = 10 μF. Find f_r, Q and bandwidth.

$f_r = 1/(2\pi\sqrt{(0.1 \times 10^{-5})}) = 1/(2\pi \times 10^{-3}) \approx$ **159.2 Hz**.

$Q = (1/R)\sqrt{(L/C)} = (1/10)\sqrt{(0.1/10^{-5})} = (1/10)(100) =$ **10**.

$BW = f_r/Q = 159.2/10 \approx$ **15.9 Hz**.

10.2.9 Three-Phase Basics

Three EMFs equal in magnitude, displaced **120°** apart. Advantages over 1-φ: more power per kg of conductor, constant total power, self-starting motors, rotating magnetic field.

Connection	Line–phase voltage	Line–phase current
Star (Y)	$V_L = \sqrt{3}\ V_{ph}$	$I_L = I_{ph}$
Delta (Δ)	$V_L = V_{ph}$	$I_L = \sqrt{3}\ I_{ph}$

> 3-φ power: $P = \sqrt{3}\ V_L\ I_L \cos\varphi = 3\ V_{ph}\ I_{ph} \cos\varphi$ (both connections)

- Star gives a **neutral** point → 3-phase 4-wire, two voltages available (e.g., 400 V line / 230 V phase; $400/\sqrt{3} = 230$).
- Phase sequence R-Y-B; reversing any two lines reverses a 3-φ motor.
- In balanced star, neutral current = 0.

Worked Example 16. A balanced 3-φ star load draws 10 A per phase at 0.8 pf from a 400 V (line) supply. Power?

$P = \sqrt{3} \times 400 \times 10 \times 0.8 \approx$ **5542 W ≈ 5.54 kW**.

10.3 Magnetic Effect of Electric Current

10.3.1 Magnetic Field — Concept ★ (asked 2025)

Magnetic field: the region (space) around a magnet or a current-carrying conductor in which its magnetic force (effect) can be experienced by another magnet, magnetic material or moving charge. Represented by lines of force (flux lines): closed loops, N → S outside the magnet, S → N inside, never intersect, mutually repel laterally, seek the path of least reluctance.

Key quantities:

Quantity	Symbol	Unit
Magnetic flux	Φ	weber (Wb)
Flux density	$B = \Phi/A$	tesla (T) = Wb/m^2
Field strength / magnetising force	$H = NI/l$	A/m (ampere-turn per metre)
Permeability	$\mu = B/H = \mu_0\mu_r$	H/m; $\mu_0 = 4\pi \times 10^{-7}$ H/m

10.3.2 Field of Conductor, Loop, Solenoid — Rules

- **Straight conductor:** concentric circular lines; $B = \mu_0 I/(2\pi d)$. Direction by **right-hand thumb rule** (thumb = current, curled fingers = field) or Maxwell's corkscrew rule.
- **Circular loop:** at centre $B = \mu_0 I/(2r)$; N-turn coil: $B = \mu_0 NI/(2r)$. Face with anticlockwise current behaves as N-pole.
- **Solenoid:** field like a bar magnet; inside $B = \mu_0 nI$ (n = turns/metre). Grip rule: fingers = current direction, thumb = N-pole.

10.3.3 Force on a Conductor; Fleming's Rules

Force on a current-carrying conductor in a field:

> $F = BIl \sin\theta$ (newton); maximum when conductor $\perp$ field ($\theta = 90°$), zero when parallel

Force on a moving charge: $F = qvB \sin\theta$ (Lorentz force, magnetic part).

Two parallel conductors: **same-direction currents attract, opposite repel**; force/metre = $\mu_0 I_1 I_2/(2\pi d)$ — basis of the ampere's classical definition.

- **Fleming's LEFT-hand rule** → **motor** action (First finger = Field, seCond = Current, thuMb = Motion/force).
- **Fleming's RIGHT-hand rule** → **generator** (induced EMF direction).

10.3.4 Magnetic Circuits ★ (asked 2025)

A magnetic circuit is the closed path followed by magnetic flux, analogous to an electric circuit:

Electric circuit	Magnetic circuit
EMF, E (V)	MMF, $F = NI$ (ampere-turns, AT)
Current, I (A)	Flux, Φ (Wb)
Resistance, $R = \rho l/A$	Reluctance, $S = l/(\mu_0\mu_r A)$ (AT/Wb)
Conductance G	Permeance = 1/S (Wb/AT)
Conductivity σ	Permeability μ
Current density J	Flux density B
$I = E/R$	**$\Phi = MMF/S = NI/S$**
Energy is continuously dissipated	No energy dissipated in maintaining flux
Current actually flows	Flux does not "flow"; it is established

> **Φ = MMF / Reluctance** — hence for the same MMF, **if reluctance increases, flux decreases** (inverse proportion). *(Direct 2025 PYQ.)*

Series magnetic circuit: $S_{total} = S_1 + S_2 + \ldots$ (add reluctances, e.g., iron path + air gap; the air gap usually dominates since $\mu_r(air) = 1$). Parallel paths: permeances add.

Worked Example 17. An iron ring of mean length 50 cm, cross-section 4 cm^2, μ_r = 1000, is wound with 200 turns carrying 2 A. Find reluctance and flux.

$S = l/(\mu_0 \mu_r A) = 0.5 / (4\pi \times 10^{-7} \times 1000 \times 4 \times 10^{-4}) = 0.5 / (5.0265 \times 10^{-7}) \approx$ **9.95×10^5 AT/Wb**.

MMF = 200 × 2 = 400 AT. $\Phi = 400 / 9.95 \times 10^5 \approx$ **4.02×10^{-4} Wb = 0.402 mWb**.

(If a 1 mm air gap were cut, $S_{gap} = 0.001/(4\pi \times 10^{-7} \times 4 \times 10^{-4}) \approx 1.99 \times 10^6$ AT/Wb — twice the iron's reluctance — flux would fall to ≈ 0.134 mWb: increased reluctance → decreased flux.)

10.3.5 B–H Curve and Hysteresis

- **B–H (magnetisation) curve:** initially steep (easy domain alignment), then knee, then **saturation** (B nearly constant despite rising H). μ is not constant for ferromagnetics.
- **Hysteresis:** B lags behind H when a magnetic material is cyclically magnetised. Loop intercepts: **residual magnetism / retentivity** (B at H = 0) and **coercive force / coercivity** (reverse H needed to make B = 0).
- **Hysteresis loss** ∝ loop area; Steinmetz: $P_h = \eta B_{max}^{1.6} f V$ (watts).
- **Hard** magnetic materials (large loop — steel, Alnico): permanent magnets. **Soft** (thin loop — silicon steel, soft iron, ferrites): transformer/machine cores. **Eddy-current loss** $P_e \propto B_{max}^2 f^2 t^2$ — reduced by **laminating** the core.

10.3.6 Electromagnetic Induction

Faraday's laws:

1. Whenever the flux linking a circuit changes, an EMF is induced (lasts only while the change lasts).
2. Magnitude of induced EMF = rate of change of flux linkages:

> $e = -N \, d\Phi/dt$

Lenz's law: the induced EMF/current always **opposes the cause** producing it (the minus sign) — a consequence of conservation of energy.

Dynamically induced EMF — conductor moves relative to the field: $e = Blv \sin\theta$ (generators). **Statically induced EMF** — conductor stationary, flux changes (transformers):

- *Self-induced:* in the same coil, $e = -L \, di/dt$.
- *Mutually induced:* in a neighbouring coil, $e = -M \, di/dt$.

Self-inductance: $L = N\Phi/I = N^2/S = \mu_0 \mu_r N^2 A/l$ (henry). A circuit has L = 1 H if 1 A/s change induces 1 V.

Mutual inductance: $M = N_2\Phi_{12}/I_1$; **coefficient of coupling**

> $k = M/\sqrt{(L_1 L_2)}$, $0 \le k \le 1$ (k = 1: perfect coupling; k small: loosely coupled)

Series combination of coupled coils: $L = L_1 + L_2 \pm 2M$ (+ aiding, − opposing).

Energy stored in inductor: $W = \frac{1}{2}LI^2$ (in the magnetic field); in capacitor: $W = \frac{1}{2}CV^2$ (electric field). Energy density of magnetic field = $B^2/2\mu$.

Worked Example 18. The flux linking a 200-turn coil changes from 4 mWb to 1 mWb in 0.1 s. Induced EMF?

$e = N \Delta\Phi/\Delta t = 200 \times (3 \times 10^{-3})/0.1 =$ **6 V**.

Worked Example 19. Two coils L_1 = 8 H, L_2 = 2 H have M = 3 H. Coefficient of coupling?

$k = 3/\sqrt{(8 \times 2)} = 3/4 =$ **0.75**.

10.4 Thermal Effect of Electric Current

10.4.1 Joule's Law of Heating

When current I flows through resistance R for time t:

> $H = I^2Rt$ joules $= I^2Rt/4.186 \approx 0.24\ I^2Rt$ calories

Heat $\propto I^2$, $\propto R$, $\propto t$. This is why the effect is called **Joule heating / ohmic heating**; it is **irreversible** and independent of current direction (unlike Peltier effect).

Worked Example 20. A 230 V, 1 kW heater runs for 30 minutes. Heat produced and energy consumed?

Energy = $Pt = 1000 \times 1800 = 1.8 \times 10^6$ J = **1800 kJ = 0.5 kWh (unit)**.

Element resistance $R = V^2/P = 230^2/1000$ = **52.9 Ω**.

10.4.2 Applications of Heating Effect

Application	Key points
Electric heater / iron / geyser	**Nichrome** element — high resistivity, high melting point, doesn't oxidise
Incandescent lamp	**Tungsten** filament (m.p. ≈ 3400 °C), argon/nitrogen filling to retard evaporation; only ~5–10% of energy → light
Electric welding, furnaces	Arc/resistance heating
Fuse	Protective device — melts and breaks the circuit on overload/short circuit

10.4.3 Fuses — Ratings & Fusing Factor

- Fuse wire: low melting point, adequate conductivity — usually **tin–lead alloy** (63% Sn/37% Pb) or tinned copper for small ratings; placed **in series** with the **live (phase)** wire, never in neutral or earth.
- **Current rating** — maximum current the fuse carries continuously without melting.
- **Fusing current** — minimum current at which it melts.

> **Fusing factor = minimum fusing current / current rating** (always > 1; rewirable ≈ 1.4–2)

- Fusing current for round wire (Preece): I_f = k d^{3/2} (d = wire diameter).
- HRC (high rupturing capacity) fuse: cartridge with silver element in quartz-sand filling — high breaking capacity, no ageing, accurate; drawback: must be replaced after operation, cost.
- Fuse protects against **short circuit / sustained overload**; it cannot protect a person against shock (too slow, too high threshold) — that is the job of ELCB/RCCB (§10.7.9).

10.5 Chemical Effect of Electric Current

10.5.1 Electrolysis

Passage of direct current through an **electrolyte** (acid/base/salt solution or molten salt) decomposes it — **electrolysis**. Current in electrolytes is carried by **ions**: cations (+) migrate to the **cathode** (− electrode), anions (−) to the **anode** (+). The vessel + electrodes + electrolyte = voltameter (electrolytic cell). Electrolysis requires **DC** (AC would merely reverse deposition each half cycle).

10.5.2 Faraday's Laws of Electrolysis

First law: mass liberated at an electrode ∝ quantity of charge passed:

> m = Z I t (Z = electrochemical equivalent, kg/C; ECE of Cu ≈ 0.3294 mg/C, of Ag ≈ 1.118 mg/C)

Second law: for the same charge through different electrolytes, masses liberated ∝ their **chemical equivalent weights** (E = atomic weight / valency); $m_1/m_2 = E_1/E_2$.

> One faraday, F ≈ 96 500 C, deposits one gram-equivalent; Z = E/F.

Worked Example 21. How much copper is deposited when 10 A flows for 30 minutes? (Z_Cu = 0.3294 mg/C)

m = ZIt = 0.3294×10^{-3} g/C × 10 × 1800 = 0.3294 × 18 = **5.93 g**.

Worked Example 22. Time to deposit 2.68 g of silver with 2 A? (Z_Ag = 1.118 mg/C)

t = m/(ZI) = 2.68 / (1.118×10^{-3} × 2) = 2.68/0.002236 ≈ 1198.6 s ≈ **20 minutes**.

10.5.3 Electroplating & Other Applications

- **Electroplating:** article to be plated = **cathode**; plating metal = **anode**; electrolyte = salt of the plating metal (e.g., $CuSO_4$ for copper, $AgNO_3$/double cyanide for silver). Purpose: corrosion protection, appearance, wear resistance.
- **Electro-refining** of metals (impure Cu anode → pure Cu cathode), **electrotyping**, **anodising** (protective oxide on aluminium — here Al is the anode), **electro-extraction** of Al/Na/Mg, manufacture of chemicals (caustic soda, chlorine).

10.5.4 Cells and Batteries

Type	Examples	Note
Primary (not rechargeable)	Dry Leclanché cell (Zn–C, EMF ≈ 1.5 V), mercury cell (1.35 V)	Chemical action irreversible
Secondary (rechargeable/storage)	**Lead–acid** (2 V/cell), Ni–Cd (1.2 V), Ni–Fe, Li-ion (3.6–3.7 V)	Action reversible; charged by DC

Lead–acid cell essentials: positive plate PbO_2 (chocolate brown), negative plate spongy Pb (grey), electrolyte dilute H_2SO_4. On **discharge**, both plates → $PbSO_4$ and acid becomes weaker (specific gravity falls from ≈ 1.28 to ≈ 1.18); charging reverses this. State of charge is best checked with a **hydrometer** (specific gravity). Capacity in **ampere-hours (Ah)**; efficiency: Ah efficiency ≈ 90%, Wh efficiency ≈ 75%. Cells in series → voltages add (same Ah); in parallel → capacities add.

- **Battery ratings/terms:** trickle charging (continuous small charge to offset self-discharge), sulphation (hard $PbSO_4$ from leaving the cell discharged — reduces capacity), gassing near full charge (H_2 + O_2 — keep flames away, ventilate).

10.6 Protection & Relays ★ (asked 2025)

(The 2025 CBT-2 asked two relay questions under this unit, so this section is exam-critical.)

10.6.1 Purpose of Protection

Power-system protection detects abnormal conditions (short circuits, earth faults, overloads) and **isolates only the faulty section** quickly, so that: equipment damage is limited, system stability is preserved, healthy sections keep supplying, and personnel are safe. Basic chain: **CT/PT (sense) → relay (decide) → circuit breaker (act)**. The relay is the *brain*, the breaker the *muscle*.

10.6.2 Desirable Qualities of Protective Relaying ★

Quality	Definition
Selectivity (discrimination) ★	Ability of the protection to **operate only for faults within its own designated (protected) zone** and to remain stable (inoperative) for faults outside that zone or for normal load conditions — only the faulty element is tripped, the rest of the system stays in service. *(Direct 2025 PYQ.)*
Sensitivity	Ability to operate for the **smallest** value of actuating quantity (minimum fault current) it is required to respond to
Speed	Fast operation to limit damage and maintain stability (fault clearing time = relay time + breaker time)
Reliability	Assurance that it will operate when required (dependability) and not operate when not required (security)
Simplicity & economy	Minimum equipment/complexity for the protection required

Backup protection operates if the primary protection fails — with an intentional **time delay** for coordination (time grading), which is itself a method of achieving selectivity (time grading, current grading, unit/differential protection).

10.6.3 Relay Types — One-Line Working

Relay	Actuating principle
Overcurrent	Operates when current exceeds a pick-up value. Characteristics: instantaneous, definite-time, IDMT (inverse definite minimum time — most common for feeders)
Directional	Responds to fault power flow in one specified direction only
Differential	Compares currents entering and leaving the protected equipment; operates when the **vector difference** exceeds a set value — a **unit protection** (inherently selective), standard for transformers/generators/busbars (Merz-Price scheme)
Distance (impedance) relay ★	**Measures the ratio V/I = impedance** at the relay point; since line impedance is proportional to length, the measured Z is **proportional to the distance to the fault** — relay operates when Z seen < set value (fault inside reach). Used for transmission-line protection in stepped zones (Zone 1 ≈ 80% instantaneous, Zone 2, Zone 3 time-delayed). Variants: impedance (circle on R–X diagram), **reactance**, and **mho** (admittance — inherently directional, circle through origin). *(Direct 2025 PYQ.)*
Buchholz relay	**Gas-actuated** relay in the pipe between a transformer's tank and conservator; slow gas evolution from incipient faults tilts the upper float → **alarm**; violent oil surge from severe internal faults strikes the lower flap → **trip**. Only for **oil-immersed transformers with conservator** (typically > 500 kVA).
Thermal relay	Bimetallic element heated by load current — overload protection of motors
Earth-fault relay	Operates on residual (zero-sequence) current during earth faults

10.6.4 Fuses vs Relays (+ Breaker)

Aspect	Fuse	Relay + circuit breaker
Function	Detects **and** interrupts itself	Relay detects; breaker interrupts
After operation	Element destroyed — replace/rewire	Simply reclose the breaker
Characteristic	Fixed inverse (melting)	Adjustable settings & time grading
Cost / use	Cheap — LV circuits, domestic	Costly — MV/HV systems, precise coordination
Speed on heavy short circuit	Extremely fast (current-limiting HRC)	Relay + breaker time (a few cycles)

10.7 Earthing — Installation, Maintenance, Testing

10.7.1 Purpose of Earthing

Connecting non-current-carrying metal parts (motor frames, appliance bodies, conduit, poles) and/or system neutral to the general mass of earth through a low-resistance path so that:

- Leakage/fault current flows safely to earth, keeping exposed metal at (nearly) earth potential → **protects persons from shock**;
- The large earth-fault current promptly operates fuses/relays/ELCB → **automatic disconnection**;
- Voltages are stabilised (neutral earthing), lightning/static discharges are safely dissipated;
- Equipment and buildings are protected from damage/fire.

Two categories: **system (neutral) earthing** — earthing the supply neutral; **equipment earthing** — earthing metallic bodies.

10.7.2 Definitions (per IS 3043 usage)

Term	Definition
Earth	The general mass of earth, taken as zero (reference) potential
Earthing (grounding)	Connecting a part to earth via conductors of negligible impedance
Earth electrode	Metal plate/pipe/rod embedded in soil making effective contact with earth
Earth continuity conductor (ECC)	The conductor connecting equipment bodies/socket earth terminals to the earthing lead — must be continuous, low resistance (its resistance from any point to the electrode should not exceed 1 Ω)
Earthing lead	The final conductor joining the earth electrode to the ECC/earth bus
Earth resistance	Total resistance between the electrode and the general mass of earth (electrode + contact + surrounding-soil resistance; the soil immediately around the electrode contributes most)

10.7.3 Plate Earthing (IS 3043)

- **Electrode:** GI (galvanised iron) plate **60 cm × 60 cm × 6.3 mm**, or **copper plate 60 cm × 60 cm × 3.15 mm**, buried **vertically** with its top not less than **3 m below ground level** (i.e., at ≈ 3 m depth), preferably in moist soil.
- The pit around the plate is filled with **alternate layers of charcoal and common salt** (each ≈ 15 cm)— charcoal retains moisture and salt lowers soil resistivity, both reducing earth resistance.

- Earthing lead (GI wire/strip, or copper with copper plate) is bolted/welded to the plate with nut-bolt-washer of the same metal and runs to ground level inside a **GI pipe** (typically 12.7 mm) for mechanical protection.
- A **funnel with wire mesh** connected to a ≈ 19 mm watering pipe is provided at the top; the arrangement is covered by a **cement-concrete inspection chamber (≈ 30 cm × 30 cm) with a cast-iron cover** for watering and periodic inspection.
- Used where soil is such that greater electrode area is needed (rock strata prevented deep pipes) and for high-capacity installations (substations, generating stations). Costlier; drying of deep soil is a drawback if not watered.

10.7.4 Pipe Earthing (most common)

- **Electrode:** GI pipe of **38 mm (≈ 40 mm) internal diameter, 2.5 m (≈ 2.0–2.5 m) length**, with the pipe placed **vertically** in a permanently wet soil layer; overall pit depth ≈ 3.75 m (per traditional practice / older IE-rule figures). The lower portion of the pipe is **perforated (holes ~12 mm)** to keep contact with moist surrounding soil.
- Surrounded, like plate earthing, by **alternate 15 cm layers of charcoal and salt** for the lowest ~2 m.
- A **19 mm funnel** with mesh at the top receives water (3–4 buckets in dry season); earth wire is clamped to the pipe top with a through-bolt and taken out in a 12.7 mm GI protection pipe; a masonry inspection chamber with CI cover finishes the top.
- Cheapest and most common method for domestic/small installations; can carry larger leakage current than an equivalent plate (larger effective surface in moist strata).

10.7.5 Rod and Strip Earthing; Other Methods

- **Rod earthing:** 12.5–16 mm dia copper rod or 16–25 mm GI/steel rod, ≥ 2.5 m long, **driven vertically** into the ground (hammered/screwed, extensible sections for deeper moisture). Cheapest — no excavation; suits sandy soil.
- **Strip/wire earthing:** GI strip (≥ 25 mm × 1.6 mm) or copper strip (25 mm × 1.6 mm) / round conductor buried **horizontally** in trenches ≥ 0.5 m deep, total length ≥ 15 m; used in **rocky terrain** where deep driving is impossible; used at transmission-line towers (counterpoise earthing).
- **Earthing through water mains** (older practice — now avoided due to non-metallic pipe sections).

10.7.6 Earthing of Installations and Appliances

- All medium-voltage (230/400 V) installations: metal parts earthed by **two separate and distinct earth connections** for transformer neutrals and important equipment (redundancy).
- 3-pin plugs/sockets: the **larger, longer top pin = earth**; appliance body wired to it via the 3-core flexible cord's green/yellow-green core.
- Conduits, switchboards, motor frames, fan hooks (via 2 earthing points for motors above certain rating), water heaters etc. all bonded to the ECC.
- **Earth wire size:** ECC not less than half the cross-section of the largest current-carrying conductor (subject to a minimum, e.g., 14 SWG copper); never less than 2.9 mm^2 (traditional rule-of-thumb; exact sizes per IS 3043 tables).
- The **neutral must never be used as earth** for appliance bodies (a broken neutral would make every body live at full phase voltage).

10.7.7 Permissible Earth-Resistance Values & Soil Resistivity

Installation	Acceptable earth resistance
Large power stations	**≤ 0.5 Ω**
Major sub-stations / EHT stations	**≤ 1 Ω**
Small sub-stations	≤ 2 Ω
Industrial premises / distribution transformers	≤ 1–2 Ω
Domestic installations (in general)	as low as practicable, commonly taken **≤ 8 Ω** (5–8 Ω range typical)
Tower footing / LT poles	≤ 10 Ω (ideally)

Factors affecting soil resistivity / earth resistance:

1. **Moisture content** — dominant factor; resistivity rises steeply below ~15–20% moisture;
2. **Type of soil** — wet marshy soil lowest; clay/loam low; sand, gravel, **rock highest**;
3. **Dissolved salts/chemical composition** — salt, charcoal, chemical treatment (bentonite) lower it;
4. **Temperature** — resistivity rises sharply near/below freezing;
5. **Depth of electrode** (deeper → moister strata → lower R), electrode size/surface area, and number of parallel electrodes (electrodes in parallel, spaced apart ≥ their length, reduce net resistance);
6. Seasonal variation — hence testing is done in the **driest season** for the worst-case value.

10.7.8 Maintenance and Testing of Earthing

Maintenance:

- **Watering** the earth pit regularly through the funnel, especially in summer (3–4 buckets); keeping the charcoal-salt bed effective (replenish periodically).
- Inspecting the chamber: cleaning, checking tightness and re-greasing of the clamp/bolt connections, checking for corrosion of lead and electrode, verifying continuity of the ECC.
- **Periodic measurement** of earth resistance (typically at least once a year, in the dry season) and recording it; if above the permissible limit — water/treat the soil, or add parallel electrodes.

Measurement — Earth tester (earth megger) and Fall-of-Potential 3-point method:

- The earth tester is a **megger-type instrument with a hand-driven generator (or electronic source) plus a current reverser/rectifier**, so that the earth is tested with (effectively) **alternating/reversed current** — DC is not used, to avoid **electrolytic/polarisation errors** in the soil — while the moving system still reads on a direct-reading ohm scale. Terminals: E (or C1-P1) to electrode under test, P (P2) to potential spike, C (C2) to current spike.
- **Fall-of-potential method:** current is circulated between the electrode **E** under test and a current spike **C** placed 25–50 m away; a potential spike **P** is driven at intermediate positions and V/I is plotted; with P at ≈ **62% of the E–C distance** (P in the flat "plateau" region, outside both resistance areas) the reading gives the true earth resistance **R = V/I**. Readings are repeated with P shifted a few metres either way — a nearly constant value confirms correct spacing.
- ECC continuity and insulation-resistance tests of the installation are done with the ordinary **megger** (insulation tester, 500 V for LV installations; IR ≥ 1 MΩ desirable).

10.7.9 Neutral vs Earth; ELCB / RCCB

Neutral	Earth
Current-carrying return conductor of the supply (star point of transformer)	Non-current-carrying safety connection to ground mass
Carries load current in normal operation	Carries current **only during a fault/leakage**
Potential near 0 V but can rise if unbalanced/broken	Held at true ground potential
Sized for full load current	Sized per earthing rules (≥ half phase conductor typ.)

Neutral is earthed **at the substation/transformer** (system earthing); at the consumer's premises, neutral and earth are kept **separate**.

ELCB / RCCB (Residual Current Circuit Breaker / Residual Current Device):

- **Current-operated ELCB = RCCB/RCD** (the modern device): phase and neutral both pass through a **toroidal core-balance CT**. Normally $I_{phase} = I_{neutral}$ → net flux = 0. If current leaks to earth (through a person or insulation fault), $I_{phase} \neq I_{neutral}$; the **residual (difference) current** induces an EMF in the CT's secondary which trips the breaker within milliseconds.
- Ratings: sensitivity 30 mA (personnel/shock protection — domestic standard), 100/300 mA (fire/equipment protection); trip time ~ 30 ms class.

- (Older) **voltage-operated ELCB:** trip coil between equipment body and earth; trips when body-to-earth voltage exceeds ~50 V — obsolete because it needs a sound earth connection and misses faults bypassing it.
- Note: an RCCB does **not** protect against overload/short circuit between phase and neutral (no residual current) — use it with MCBs (or use an RCBO which combines both).

10.8 PYQ Corner — Actual CBT-2 (22-04-2025) Questions Solved

PYQ 1. *Which quality of a protective relay ensures that it operates only for faults within its designated zone of protection?*

Answer: Selectivity (discrimination).

Explanation: Selectivity is the ability of protection to select and disconnect **only the faulty part**, operating for faults **inside its own protected zone** and remaining stable for external (through) faults and normal load. Sensitivity concerns the minimum operating quantity, speed the operating time, reliability the assurance of correct operation.

PYQ 2. *A distance relay operates on the principle of —*

Answer: measurement of the impedance (ratio of voltage to current, Z = V/I) at the relay location.

Explanation: The relay is fed V (via PT) and I (via CT). For a fault on a transmission line, V/I equals the line impedance up to the fault, which is **directly proportional to the distance of the fault** from the relay. If the measured impedance is less than the set (reach) impedance, the fault is inside the zone and the relay trips. Hence the names distance/impedance relay; variants: reactance and mho relays.

PYQ 3. *In a magnetic circuit, if the reluctance is increased while the MMF is kept the same, the flux will —*

Answer: decrease.

Explanation: Ohm's law of the magnetic circuit: **Φ = MMF/S**. With MMF constant, $\Phi \propto 1/S$. Example: MMF = 400 AT, S = 1×10^6 AT/Wb → Φ = 0.4 mWb; if S doubles to 2×10^6 (e.g., an air gap is introduced), Φ halves to 0.2 mWb.

PYQ 4. *The region around a magnet in which its influence (magnetic force) can be experienced is called —*

Answer: the magnetic field.

Explanation: The magnetic field is the space surrounding a magnet or current-carrying conductor in which magnetic force acts on another magnet, ferromagnetic material or moving charge. It is mapped by lines of force which form closed loops (N → S outside the magnet) and never intersect. (Contrast: the electric field is the region where a force acts on a static charge.)

10.9 Practice MCQs (25)

Q1. Ohm's law is not applicable to —

(a) copper conductors (b) manganin resistors (c) semiconductor diodes (d) heating elements

Q2. Two resistors of 4 Ω and 12 Ω in parallel give an equivalent of —

(a) 16 Ω (b) 8 Ω (c) 3 Ω (d) 0.33 Ω

Q3. Kirchhoff's current law is based on the law of conservation of —

(a) energy (b) charge (c) momentum (d) mass

Q4. Three equal resistors R connected in delta are replaced by an equivalent star. Each star resistance is —

(a) 3R (b) R (c) R/3 (d) R/9

Q5. Maximum power is transferred to a DC load when R_L equals —

(a) zero (b) infinity (c) R_Th of the source network (d) twice R_Th

Q6. Efficiency of a source at the maximum-power-transfer condition is —

(a) 100% (b) 75% (c) 50% (d) 25%

Q7. The form factor of a sinusoidal wave is —

(a) 1.414 (b) 1.11 (c) 0.707 (d) 0.637

Q8. The RMS value of v = 100 sin ωt is —

(a) 100 V (b) 70.7 V (c) 63.7 V (d) 50 V

Q9. In a pure inductor, the current —

(a) leads the voltage by 90° (b) lags the voltage by 90° (c) is in phase with voltage (d) lags by 45°

Q10. A series RLC circuit at resonance has —

(a) maximum impedance (b) minimum current (c) unity power factor (d) zero current

Q11. The Q-factor of a series resonant circuit equals —

(a) R√(C/L) (b) (1/R)√(L/C) (c) R/(2πfL) (d) 2πfCR

Q12. In a parallel resonant (tank) circuit the line current at resonance is —

(a) maximum (b) minimum (c) zero always (d) equal to the circulating current

Q13. In a 3-phase star connection, V_line/V_phase is —

(a) 1 (b) √3 (c) 3 (d) 1/√3

Q14. A capacitor connected to DC in steady state acts as —

(a) short circuit (b) open circuit (c) pure resistance (d) inductor

Q15. The unit of MMF is —

(a) weber (b) tesla (c) ampere-turn (d) AT/Wb

Q16. If an air gap is introduced in an iron-core magnetic circuit (same MMF), the flux —

(a) increases (b) decreases (c) is unchanged (d) becomes zero

Q17. Fleming's left-hand rule is used to find the —

(a) direction of induced EMF (b) direction of force on a current-carrying conductor in a field (c) polarity of a solenoid (d) direction of flux in a core

Q18. Lenz's law is a consequence of the law of conservation of —

(a) charge (b) flux (c) energy (d) current

Q19. Two coupled coils of 4 H and 9 H have mutual inductance 3 H. The coefficient of coupling is —

(a) 1 (b) 0.75 (c) 0.5 (d) 0.33

Q20. Heat produced in a resistor is proportional to —

(a) I (b) I^1.5 (c) I^2 (d) √I

Q21. Fusing factor of a fuse is —

(a) always less than 1 (b) equal to 1 (c) always greater than 1 (d) zero

Q22. In Faraday's first law of electrolysis, the mass deposited is proportional to —

(a) voltage applied (b) quantity of electricity (charge) passed (c) resistance of electrolyte (d) temperature only

Q23. In electroplating, the article to be plated is made the —

(a) anode (b) cathode (c) electrolyte (d) separator

Q24. A Buchholz relay is used for the protection of —

(a) transmission lines (b) alternators (c) oil-immersed transformers (d) busbars

Q25. As per standard practice, the earth resistance of a large power station should not exceed about —

(a) 8 Ω (b) 5 Ω (c) 1–2 Ω (d) 0.5 Ω

Answer Key with Brief Explanations

Q	Ans	Explanation
1	(c)	Diode is a non-linear, unilateral device — V–I not proportional
2	(c)	$4\times12/(4+12) = 48/16 = 3\ \Omega$
3	(b)	Charge cannot accumulate at a node — conservation of charge
4	(c)	Delta→star with equal arms: $R_Y = R_\Delta/3$
5	(c)	Maximum power transfer: $R_L = R_{Th}$ (source internal resistance)
6	(c)	Equal power lost in $R_{Th} \Rightarrow \eta = 50\%$
7	(b)	RMS/Average = $0.707I_m/0.637I_m = 1.11$
8	(b)	$V_{rms} = 100/\sqrt{2} = 70.7$ V
9	(b)	Pure L: I lags V by 90° (CIVIL)
10	(c)	$X_L = X_C \Rightarrow Z = R$ (min), I max, $\cos\varphi = 1$
11	(b)	$Q = \omega_r L/R = (1/R)\sqrt{(L/C)}$
12	(b)	Parallel resonance: impedance max (L/CR) ⇒ line current min
13	(b)	Star: $V_L = \sqrt{3}\ V_{ph}$
14	(b)	Fully charged capacitor blocks DC — open circuit
15	(c)	MMF = NI, unit ampere-turn (AT)
16	(b)	Air gap adds large reluctance; Φ = MMF/S falls
17	(b)	Left hand = motor action (force); right hand = generator (EMF)
18	(c)	Opposition to the cause ensures work must be done — energy conservation
19	(c)	$k = M/\sqrt{(L_1L_2)} = 3/\sqrt{36} = 0.5$
20	(c)	Joule's law $H = I^2Rt$
21	(c)	Fusing current > current rating, so factor > 1 (≈1.4–2 rewirable)
22	(b)	$m = ZIt = ZQ$ — proportional to charge
23	(b)	Metal ions (cations) deposit on the cathode = article
24	(c)	Gas-operated relay between tank and conservator of oil transformers
25	(d)	Power stations ≤ 0.5 Ω; major substations ≤ 1 Ω; homes ≈ ≤ 8 Ω

10.10 Rapid-Revision Formula Sheet

DC circuits

- $Q = It$; $V = W/Q$; $P = VI = I^2R = V^2/R$; $W = Pt$; 1 kWh = 3.6×10^6 J; 1 hp = 746 W
- $R = \rho l/A$; $G = 1/R$; $R_2 = R_1[1 + \alpha(t_2 - t_1)]$
- Series: $R = \Sigma R_i$; Parallel: $1/R = \Sigma 1/R_i$; two in parallel: product/sum; n equal: nr and r/n
- Divider rules: $V_1 = V \cdot R_1/(R_1+R_2)$; $I_1 = I \cdot R_2/(R_1+R_2)$
- KCL: $\Sigma I = 0$ (charge); KVL: $\Sigma E = \Sigma IR$ (energy)
- $\Delta \rightarrow Y$: $R_1 = R_{12}R_{31}/\Sigma R_\Delta$; $Y \rightarrow \Delta$: $R_{12} = R_1 + R_2 + R_1R_2/R_3$; equal case: $R_Y = R_\Delta/3$
- Thevenin: $I_L = V_Th/(R_Th + R_L)$; Norton: $V_Th = I_N\ R_N$; MPT: $R_L = R_Th$, $P_max = V_Th^2/4R_Th$, $\eta = 50\%$

AC fundamentals

- $e = E_m \sin \omega t$; $\omega = 2\pi f$; $T = 1/f$; $f = PN/120$
- Sine: $I_rms = 0.707 I_m$; $I_av = 0.637 I_m$; FF = 1.11; PF(crest) = 1.414
- $X_L = 2\pi fL$; $X_C = 1/(2\pi fC)$; $Z_series = \sqrt{[R^2 + (X_L - X_C)^2]}$; $\cos \varphi = R/Z$
- $P = VI \cos \varphi$ (W); $Q = VI \sin \varphi$ (VAR); $S = VI$ (VA); $S^2 = P^2 + Q^2$
- Series resonance: $f_r = 1/(2\pi\sqrt{LC})$; $Z_min = R$; $Q = (1/R)\sqrt{(L/C)} = f_r/BW$
- Parallel resonance: $Z_max = L/CR$ (dynamic impedance); line current minimum
- 3-φ: star $V_L = \sqrt{3} V_ph$, $I_L = I_ph$; delta $V_L = V_ph$, $I_L = \sqrt{3} I_ph$; $P = \sqrt{3} V_L I_L \cos \varphi$

Magnetic

- $B = \Phi/A$ (T); $H = NI/l$ (A/m); $B = \mu_0 \mu_r\ H$; $\mu_0 = 4\pi \times 10^{-7}$ H/m
- $F = BIl \sin \theta$; parallel currents: same direction attract; $f/m = \mu_0 I_1 I_2/2\pi d$
- Magnetic Ohm's law: **Φ = MMF/S**; $S = l/(\mu_0 \mu_rA)$ (AT/Wb); series reluctances add
- $e = -N\ d\Phi/dt$; $e = Blv$; $e_L = -L\ di/dt$; $L = N\Phi/I = N^2/S$
- $k = M/\sqrt{(L_1L_2)}$; series-coupled $L = L_1 + L_2 \pm 2M$; $W = \frac{1}{2}LI^2$
- $P_h \propto B_max\text{^}1.6\ f$ (Steinmetz); $P_e \propto B^2f^2t^2$ (laminate to reduce)

Thermal & chemical

- $H = I^2Rt$ J = 0.24 I^2Rt cal; fusing factor = fusing current/current rating (> 1); $I_f = kd\text{^}\{3/2\}$
- $m = ZIt$; $Z = E/F$; $F \approx 96\,500$ C; $m_1/m_2 = E_1/E_2$; plating: article = cathode
- Lead–acid: 2 V/cell, H_2SO_4 sp. gr. 1.28 (charged) → 1.18 (discharged); capacity in Ah

Protection & relays

- Qualities: **Selectivity** (own zone only), Sensitivity (min operating quantity), Speed, Reliability
- Distance relay: measures $Z = V/I \propto$ fault distance; zones 1/2/3; mho = directional
- Differential: vector difference of in/out currents (unit protection); Buchholz: gas-operated, oil transformers
- Fuse in phase wire only; HRC = silver element in quartz sand

Earthing

- Plate: GI 60 × 60 cm × 6.3 mm (Cu 3.15 mm), top ≥ 3 m deep, charcoal + salt layers, funnel for watering
- Pipe: 38 mm dia × 2.5 m perforated GI pipe, ≈ 3.75 m pit, charcoal + salt — commonest, cheapest
- Earth resistance limits: power station ≤ 0.5 Ω; big substation ≤ 1 Ω; small SS ≤ 2 Ω; homes ≈ ≤ 8 Ω
- Soil resistivity ↓ with moisture, salts, temperature (above freezing), depth; rock worst
- Testing: earth tester (reversed-current megger) — fall-of-potential 3-spike method, P at ≈ 62% of E–C distance; $R = V/I$
- RCCB: core-balance CT trips on residual (phase − neutral) current; 30 mA for shock protection; does not replace MCB

- Never use neutral as earth; 3-pin plug: thick top pin = earth

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End of Chapter 10. Cross-links: magnetic materials → Ch01 (Components & Materials); measurement of R/L/C and meggers → Ch06 (Electronic Measurements).